

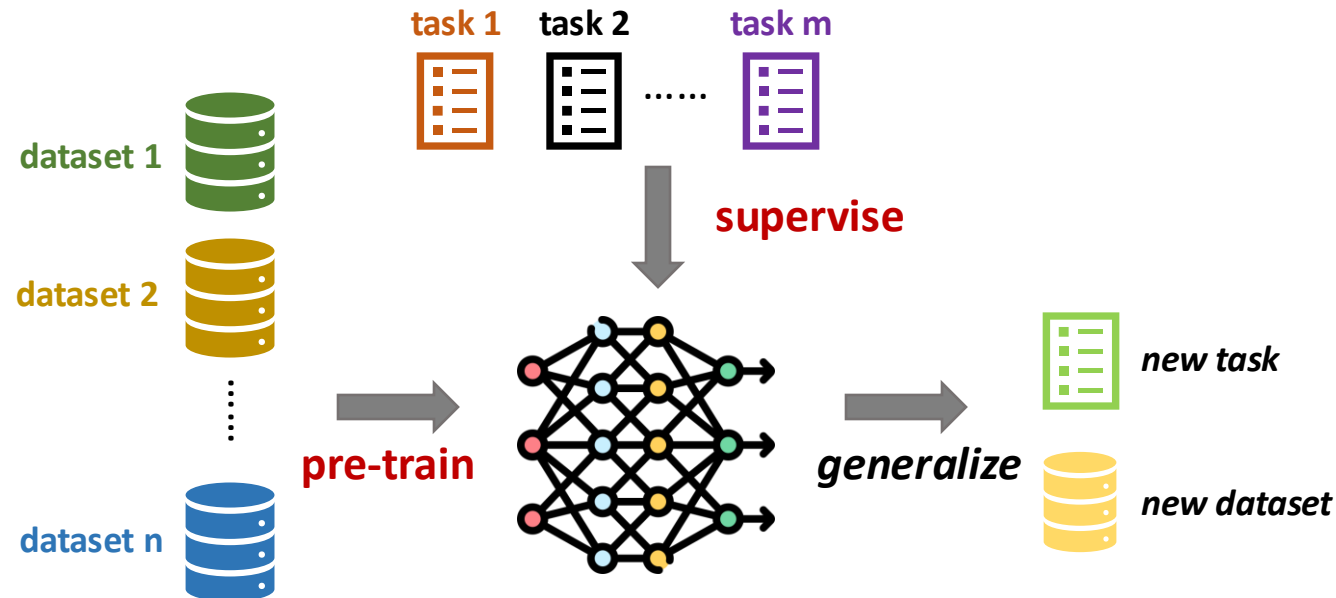
Graph Foundation Models

CPSC483: Deep Learning on Graph-Structured Data

Rex Ying

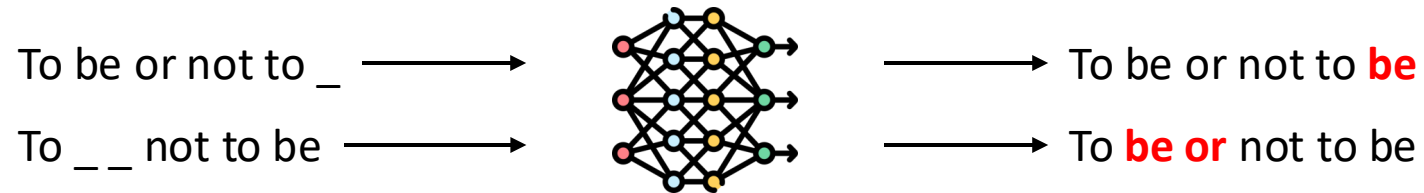
Foundation Models (1)

- Foundation Models are recently attracting interest in NLP, CV, Audio, etc.
 - Leverage extensive pre-training on diverse datasets
 - Achieve generalizable representations
 - Enable effective transfer to diverse downstream tasks
- Foundation model pre-training paradigm:

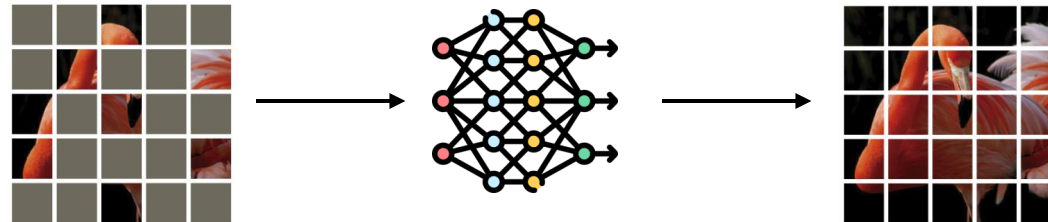


Foundation Models (2)

- Context as supervision offers generalizable representation
 - For natural language: next token prediction and masked token prediction



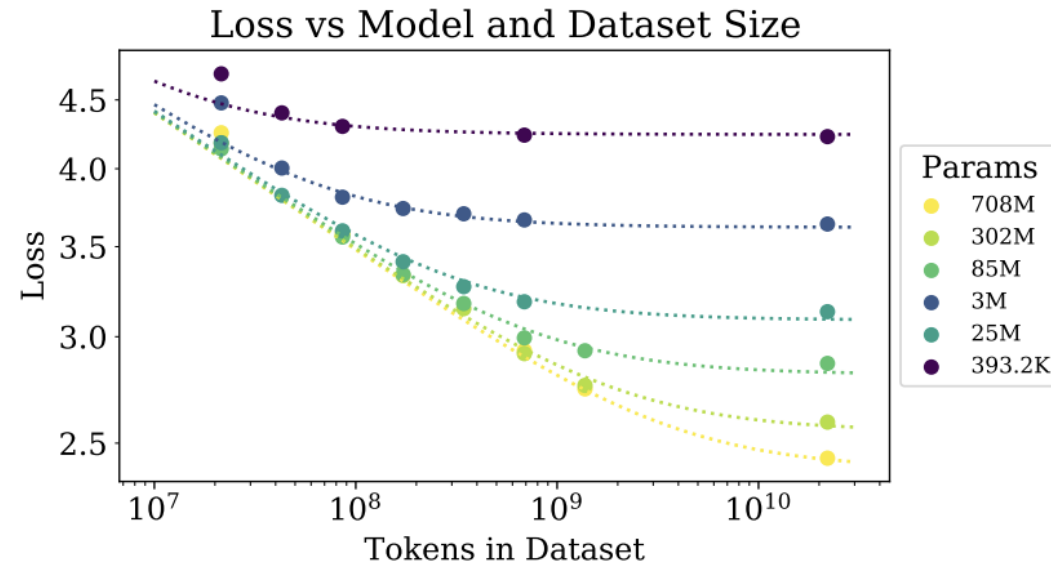
- For image: image reconstruction



Foundation Models (3)

Scaling law

- The performance can consistently be improved with larger model and dataset size



- How can we develop foundation models on **graphs**?

Graph Foundation Models

- **Graph Foundation Models for Structure Encoding**
 - **WWW2026 - Towards A Universal Graph Structural Encoder**
- **Graph Foundation Models for Relational Database**
 - **ICLR2026 - Relational Transformer: Toward Zero-Shot Foundation Models for Relational Data**
- **Graph Foundation Models on Biology**
 - **ICLR2026 - HEIST: A Graph Foundation Model for Spatial Transcriptomics and Proteomics Data**

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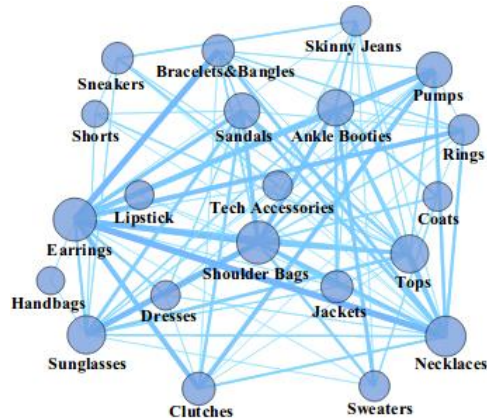
Challenges of Graph Foundation Models

- There are inherent differences in **semantic meanings** of graph features across various contexts.
- What is transferable knowledge across graph data?
- What distribution is to be modeled in graph foundation models?

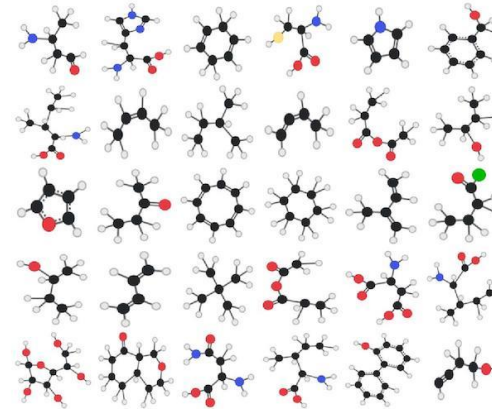
Social network



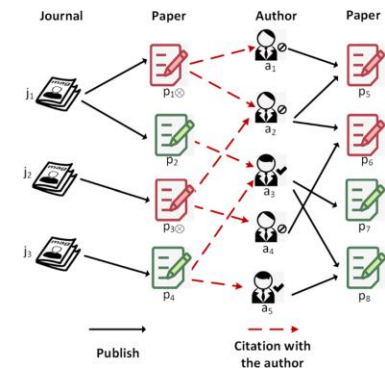
Retail product network



Molecular network



Citation network



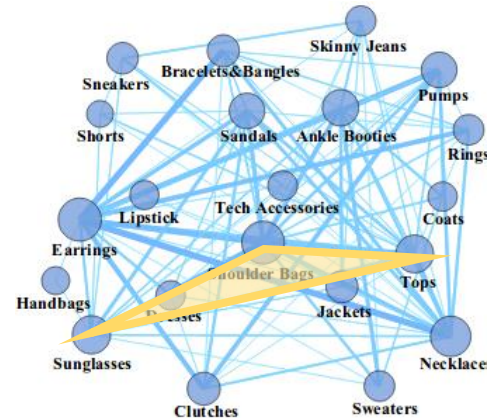
Structural Patterns as Universal Attributes

- Domain-agnostic and universal attributes: **Structural Patterns**
 - e.g., small-world property, recurring motifs, scale-free degree distributions, etc.
- **Structural Patterns** facilitate knowledge transfer across tasks and datasets.

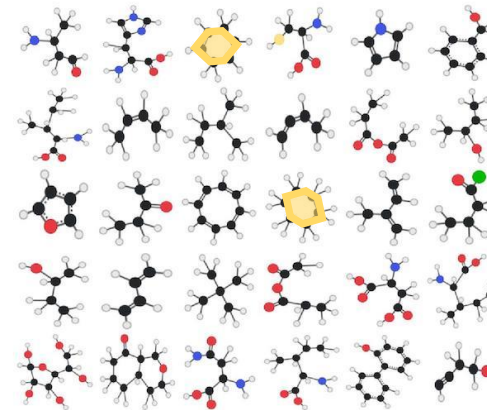
Social network



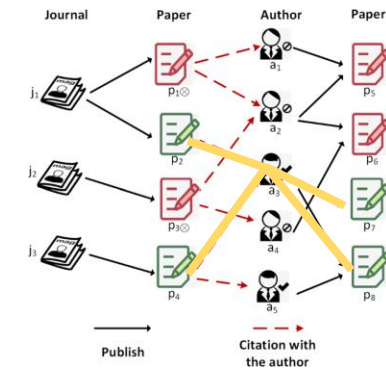
Retail product network



Molecular network



Citation network

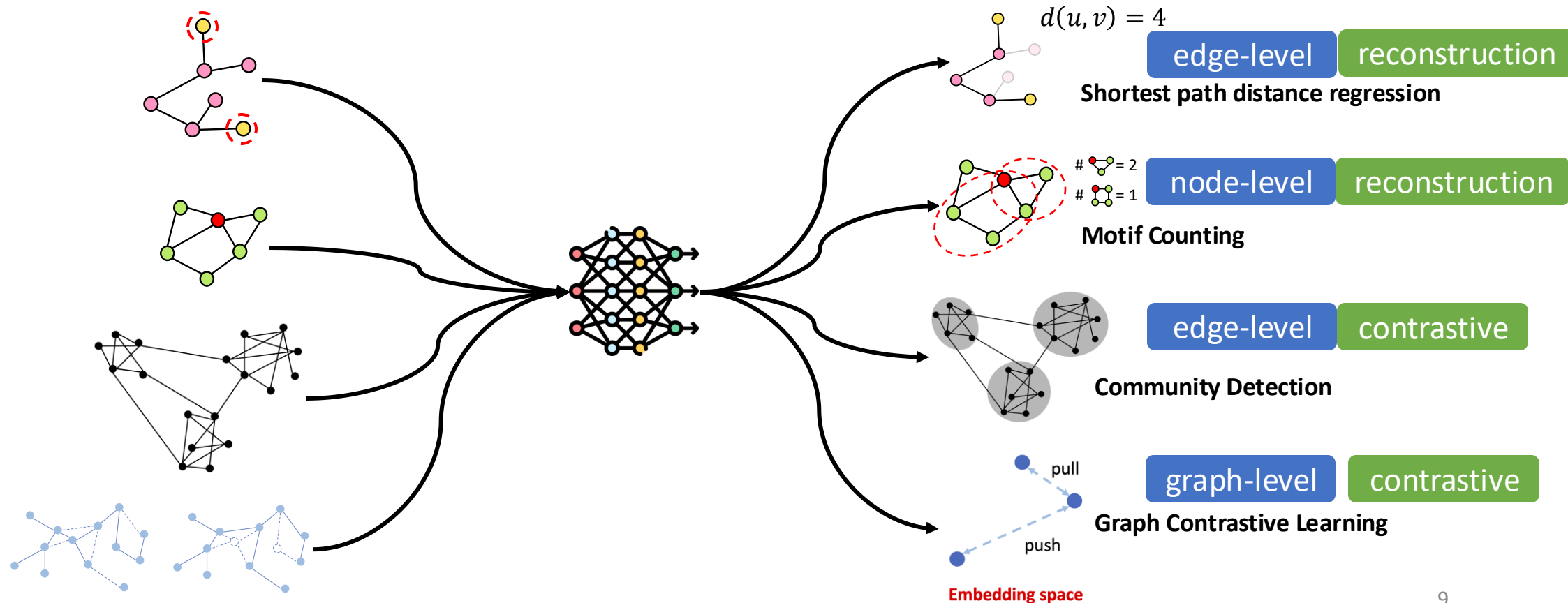


Token ↔ Node

Context ↔ Structure

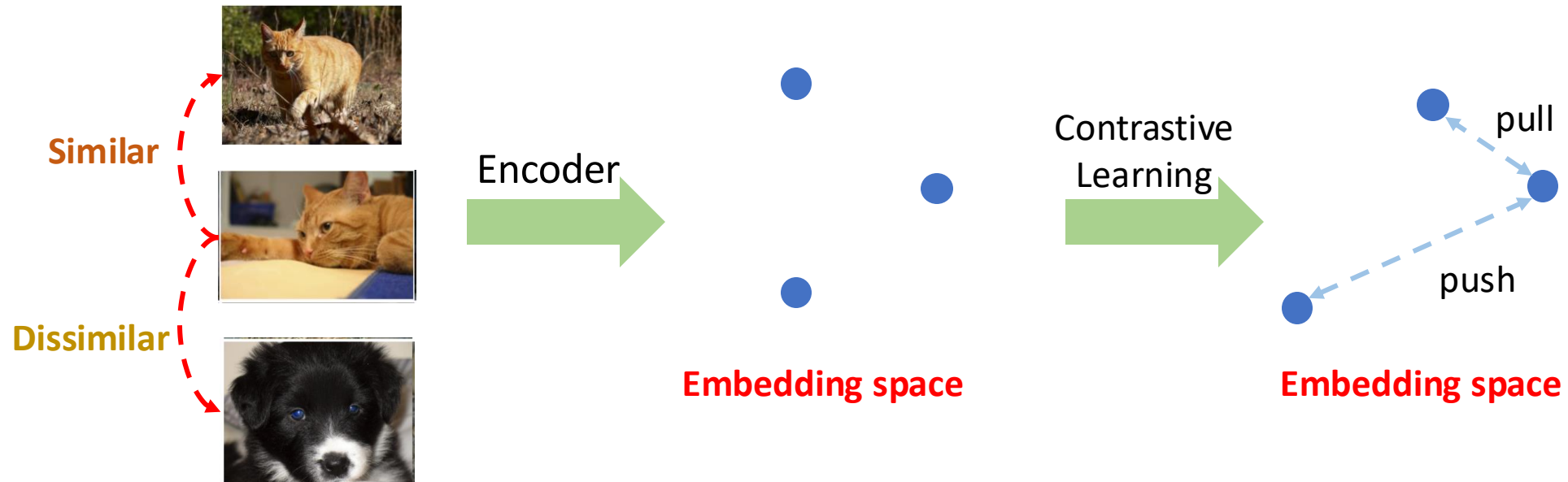
Structural Self-Supervised Tasks

- Context as supervision: we can pretrain graph foundation models by predicting graph structural features



Graph Contrastive Learning (1)

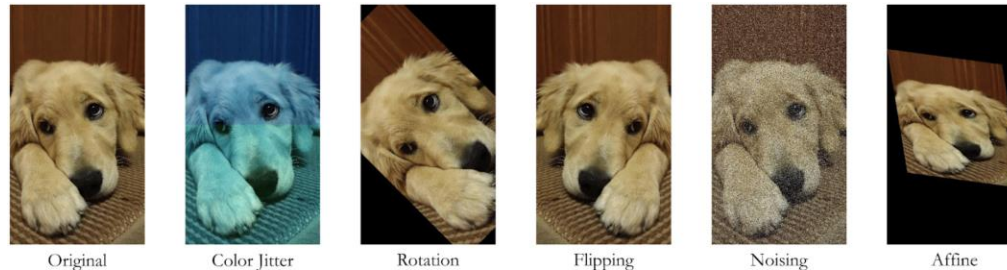
- A pretrained model should be able to exploit the **semantic features**
 - Produce similar representations for similar instances
- Contrastive learning
 - Bringing **semantically similar** instances closer while pushing **dissimilar** instances



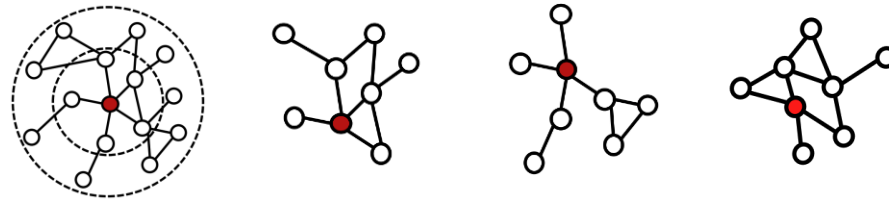
Graph Contrastive Learning (2)

- Apply data augmentation on the **anchor** to generate the **positive sample**
 - The anchor is made to undergo transformations and used as positive sample to the anchor
 - Create diverse yet semantically consistent instances
 - E.g.,

Vision([credit](#))

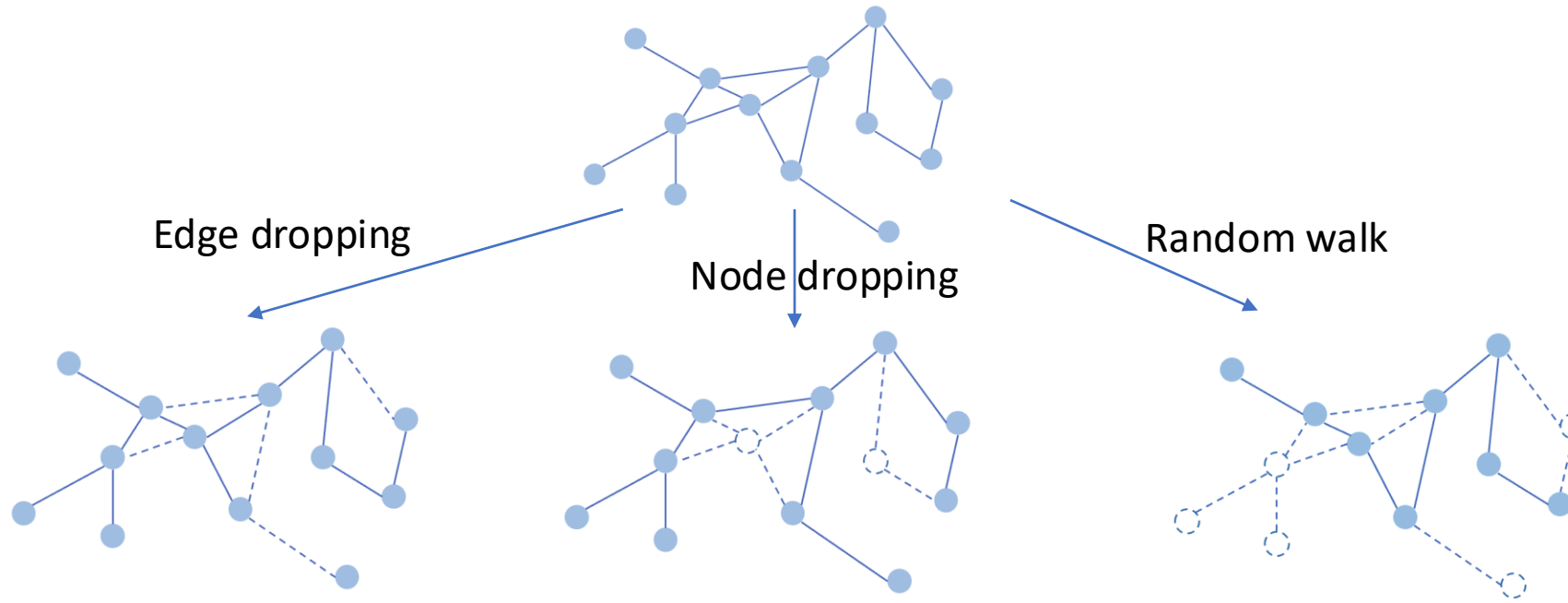


Graph([credit](#))



Graph Contrastive Learning (3)

- Some commonly-used data augmentation methods on graph
 - Edge/Node dropping: randomly dropping edges/nodes
 - Random walk: Generating sequences of nodes by randomly traversing edges



[1] Data Augmentation for Deep Graph Learning: A Survey

[2] Graph Contrastive Learning with Augmentations, NeurIPS2020

Graph Contrastive Learning (4): InfoNCE Loss

- Optimized model with InfoNCE loss

- Sample a mini-batch of instances $\{x_i\}_B$ (**anchor**) and apply data augmentation to generate **positives** $\{x_i^+\}_B$

- InfoNCE loss

$$\begin{aligned} \min - \sum_{i=1}^B \log \frac{e^{f(x_i)^T f(x_i^+)}}{\sum_j e^{f(x_i)^T f(x_j)}} \end{aligned}$$

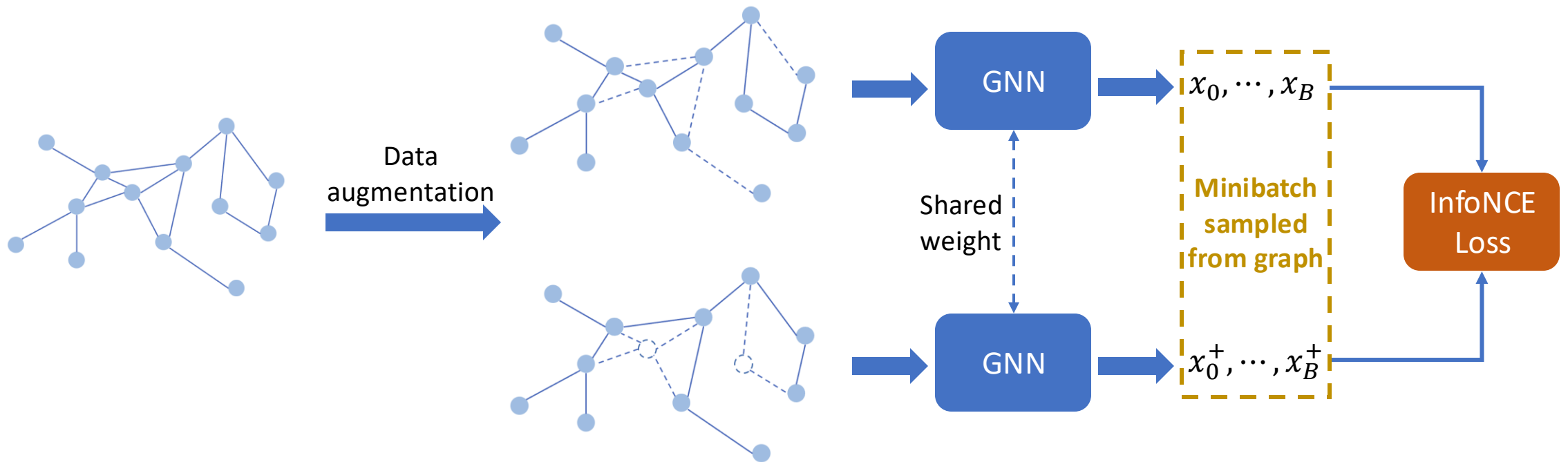
$f(\cdot)$: encoder
 x_i : anchor
 x_i^+ : positive sample

$$= \min - \sum_{i=1}^B \left(\underbrace{f(x_i)^T f(x_i^+)}_{\text{minimize distance}} - \log \underbrace{\sum_j e^{f(x_i)^T f(x_j)}}_{\text{maximize distance}} \right)$$

- One **anchor** is paired with a **positive** sample, while the remaining instances in the mini-batch serve as **negative** samples

Graph Contrastive Learning (5)

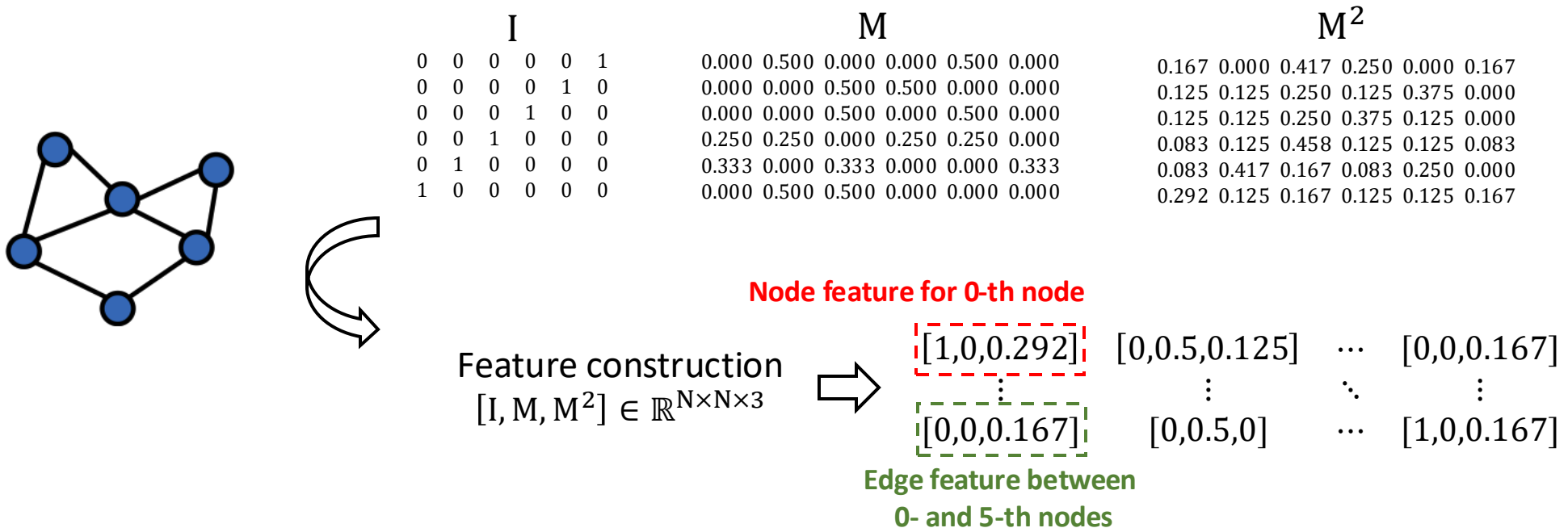
- Aligning representations between two augmented graphs[1]



[1] Graph Contrastive Learning with Augmentations, NeurIPS2020

GFSE Architecture (1)

- Construct node and edge features based on graph structure
 - A: adjacent matrix; D: degree matrix; $M = D^{-1}A$ indicates the transition matrix
 - Concatenating transition probabilities across different powers
 - Diagonal elements serve as **node features**, while off-diagonal elements serve as **edge features**

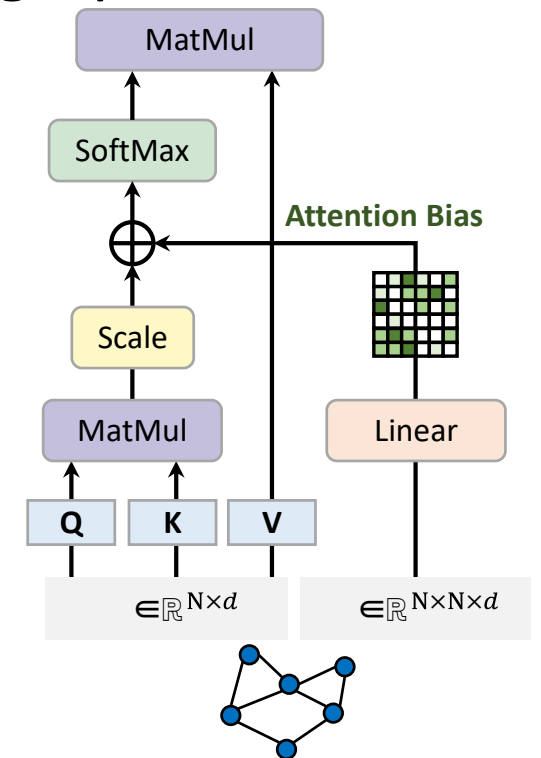


GFSE Architecture (2)

- GFSE is built on GPS^[1] layers, a hybrid approach involving both **local message passing** and **global attention** across the entire graph.
- Feature construction
 - d -dim node feature: $P_i = [I, M, M^2, \dots, M^d]_{i,i}$
 - d -dim edge feature: $R_{i,j} = [I, M, M^2, \dots, M^d]_{i,j}$
- The attention score is biased by relative encoding $\mathbf{R}$

$$a'_{i,j} = \text{SoftMax} \left(a_{i,j} + \text{Linear}(\mathbf{R}_{i,j}) \right)$$

Directly enhances the model's expressiveness



Expressive Power of GFSE

- We employ **RW-SEG-WL** to characterize the expressive power of GFSE

Definition of RW-SEG-WL: RW(d)-SEG-WL updates the node labels at the t -th iteration by $g_t(v) = \text{Hash}(\{(g_{t-1}(u), f_R(v, u)): u \in V\})$ and $g_0(v) = \text{Hash}(f_P(v))$ where $f_P(v_i) = \mathbf{P}_i \in \mathbb{R}^d$ and $f_R(v_i, v_j) = \mathbf{R}_{ij} \in \mathbb{R}^d$.

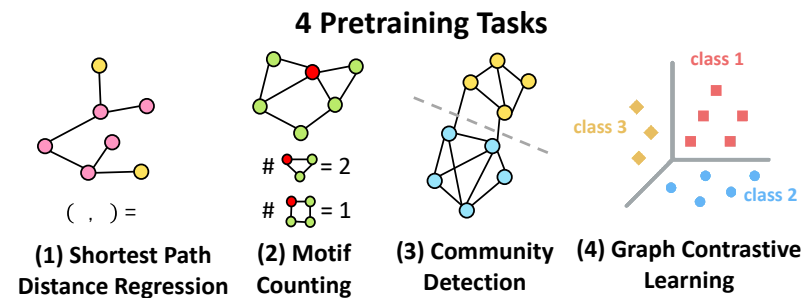
- RW-SEG-WL stands as an expressivity upper bound of GFSE
- The pre-training tasks are designed to push GFSE towards achieving the upper bound established by RW-SEG-WL

Proposition 1: RW(d)-SEG-WL is strictly more expressive than 1-WL in testing non-isomorphic graphs with $d \geq 3$.

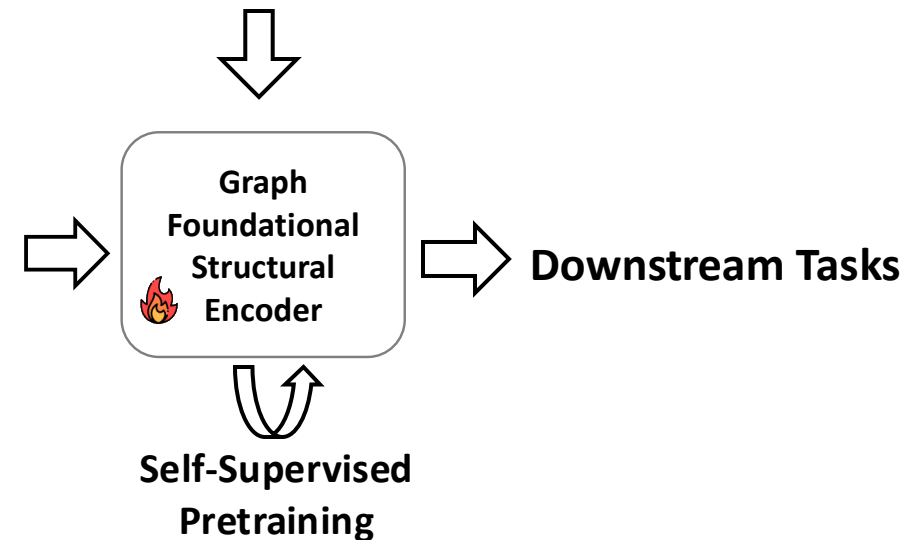
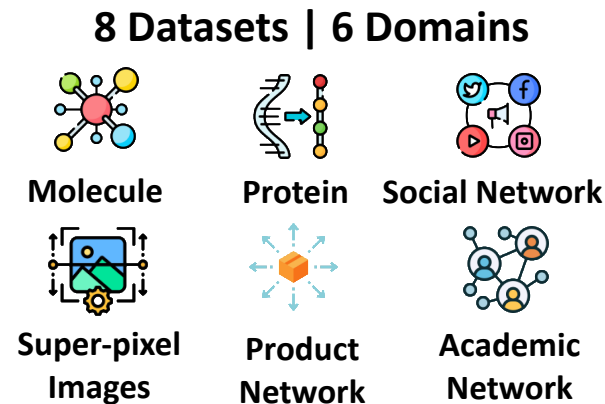
Proposition 2: There exist pairs of graphs that RW(d)-SEG-WL can distinguish, but 3-WL can not.

Graph Foundation Model Pre-training

- Graph Foundational Structural Encoder (GFSE) Pipeline

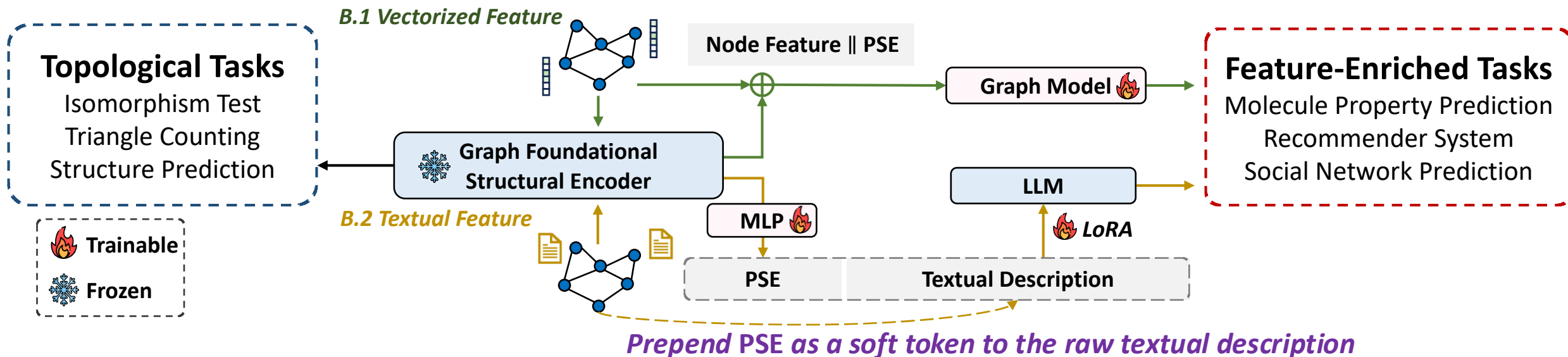


8 graph datasets from **6 different domains** to ensure the diversity in pre-training dataset.



Downstream Application

- GFSE provides expressive Positional and Structural Encoding (PSE) for graph.
- Usage: integrate into downstream graph encoders
 - MPNN and Transformers are good encoders for graphs with **vectorized features**
 - Large Language Models are good encoders for **text-attributed graphs**



Downstream Evaluation: Real-world Graphs

- The effectiveness of structural encoding typically varies across different datasets.
 - RWSE tends to be more beneficial for small molecular graph learning (e.g., MolPCBA)
- The performance gains brought by GFSE are more significant when integrating PSE with **Transformer** architecture, which lacks graph structural sensitivity

	MolPCBA AP $\uparrow$	ZINC MAE $\downarrow$	Peptides-func AP $\uparrow$	Peptides-struct MAE $\downarrow$
GCN	0.2424	0.3670	0.5930	0.3496
GCN+LapPE	0.2417	0.2052	0.6021	0.2688
GCN+RWSE	0.2438	0.1741	0.5827	0.3270
GCN+GPSE	0.1958	0.1218	0.5959	0.2710
GCN+GFSE	0.2477	0.1237	0.6131	0.2513
GIN	0.2703	0.5260	0.5498	0.3547
GIN+LapPE	0.2701	0.2203	0.5323	0.2650
GIN+RWSE	0.2781	0.1731	0.5410	0.3282
GIN+GPSE	0.2765	0.2162	0.5389	0.2581
GIN+GFSE	0.2839	0.1689	0.5532	0.2674
Transformer (TF)	0.0808	0.6943	0.4800	0.4192
TF+LapPE	0.1784	0.5101	0.6307	0.2514
TF+RWSE	0.2083	0.2193	0.6326	0.3344
TF+GPSE	0.2040	0.1883	0.6534	0.2479
TF+GFSE	0.2376	0.1548	0.6642	0.2436
GPS	0.2869	0.1182	0.6535	0.2500
GPS+LapPE	0.2939	0.1078	0.6494	0.2501
GPS+RWSE	0.2907	0.0700	0.6603	0.2739
GPS+GPSE	0.2911	0.0648	0.6688	0.2464
GPS+GFSE	0.2916	0.0613	0.6874	0.2474
GFSE Imp.(%)	32.60	76.43	2.78	42.47

Evaluation on Text-attributed Graphs

- We select three categories from Amazon e-commerce network: Cloth, Home and Sport. Each is a large-scale text-attributed graphs.
- Each node (item) is associated with its textual description.
- Task: predict the items to be connected

Table 5: Comparison with general-domain baselines

	Cloth		Home		Sport	
	Hit@1	MRR	Hit@1	MRR	Hit@1	MRR
InstructGLM	76.23	82.60	79.82	85.93	62.50	73.25
Finetuned LLaMA	74.73	82.87	78.93	86.07	62.52	75.77
+ GraphSAGE	76.22	84.16	73.74	81.66	62.26	75.36
+ GFSE (Ours)	76.84	84.68	79.85	86.77	64.79	76.24

InstructGLM: fine-tuned on graph domains without structural information infusion.

Use GraphSAGE to replace the pre-trained GFSE.
Train GraphSAGE and finetune LLaMA together

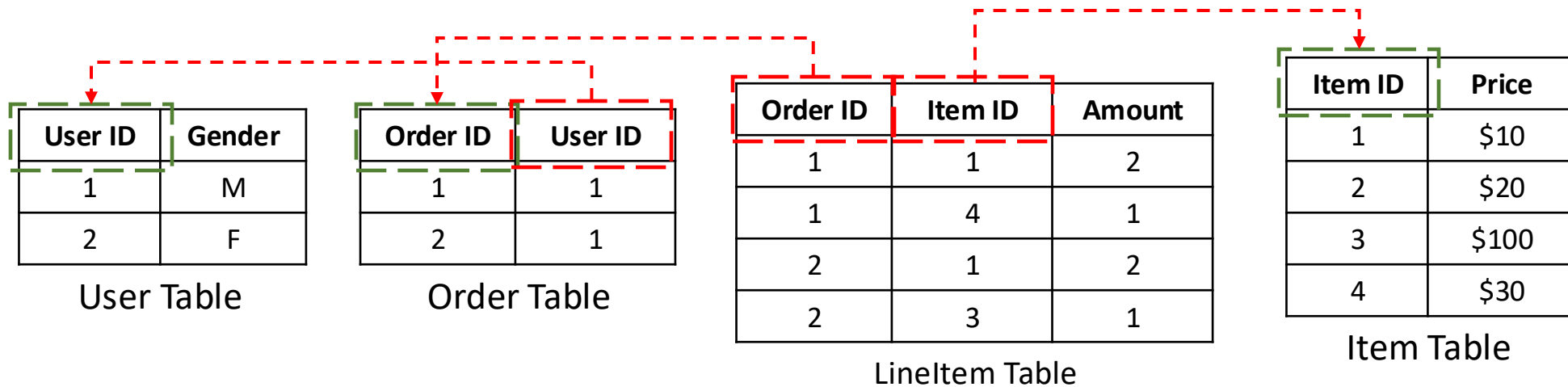
- GFSE consistently outperforms other methods across datasets.
- Cross-domain Pre-training gives benefit to high-quality PSE.

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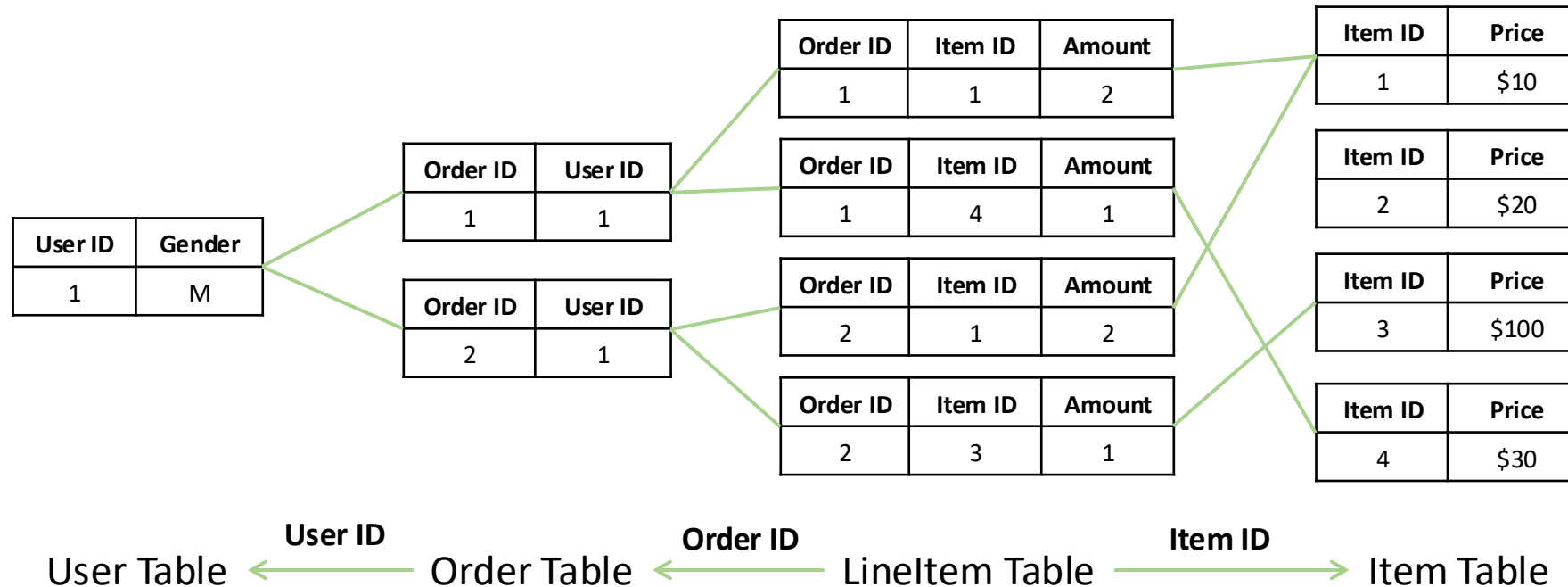
Relational Database

- A **Relational Database** is a digital repository that organizes data into structured tables consisting of rows and columns
 - Primary key (P-key): A unique identifier for a specific record within a table
 - Foreign key (F-key): A column in one table that points to the P-key of another table
- A simple 4-schema online shopping database:



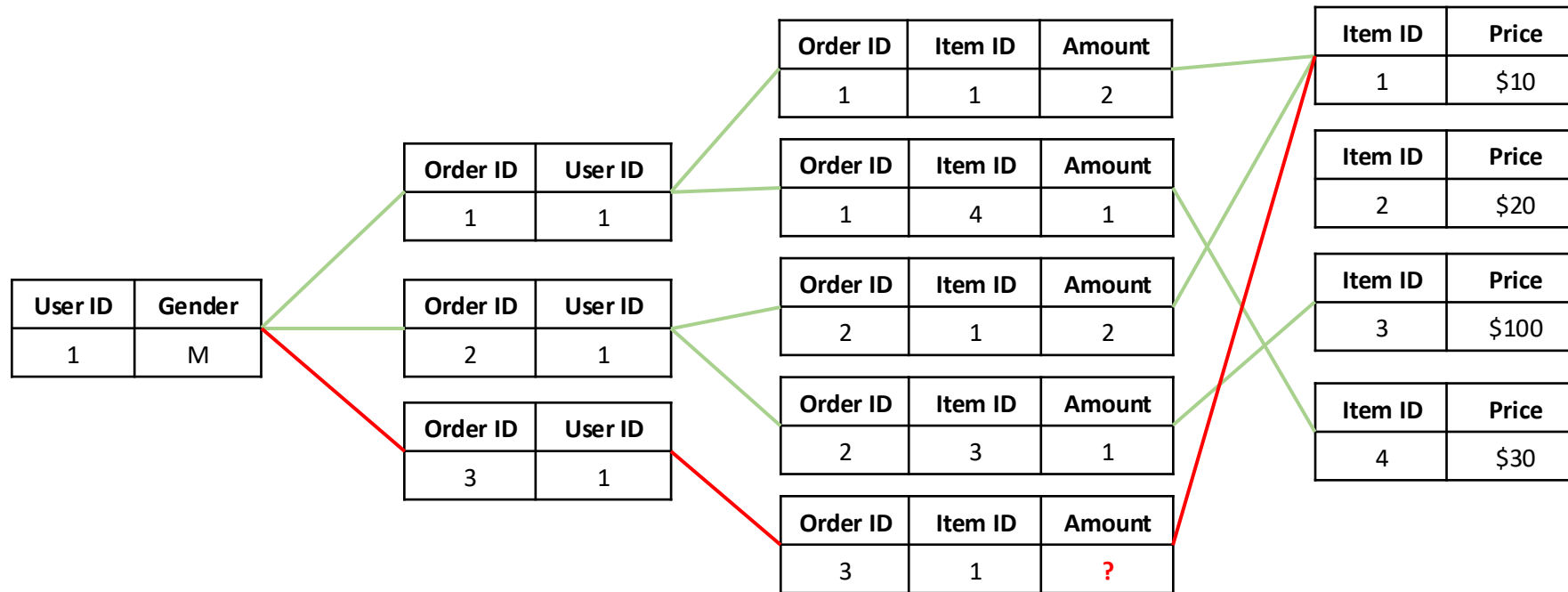
Relational Database as Graph (1)

- Relational database is a graph
 - The graph structure turns isolated data rows into a connected web of knowledge
 - Every row is a node, and the link between F-key and P-key is an edge



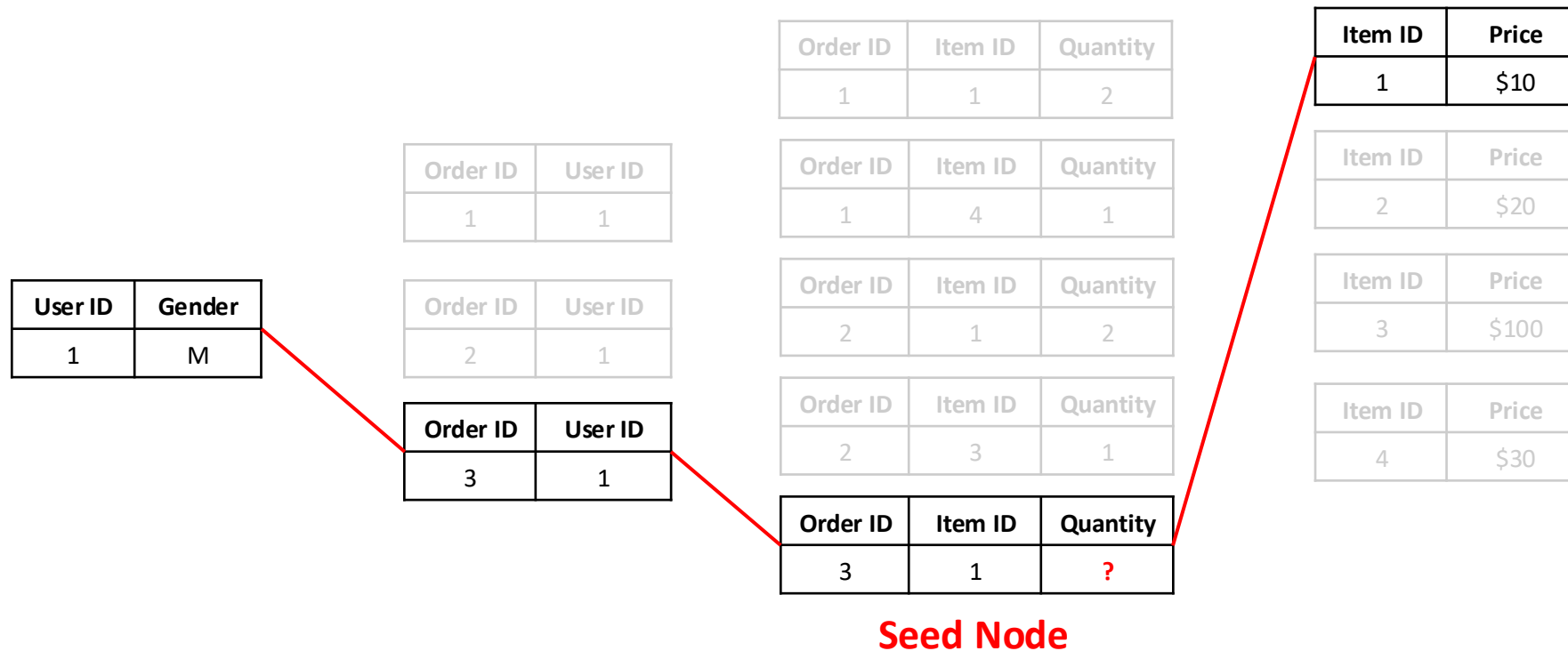
Relational Database as Graph (2)

- Predictive tasks on relational database are now graph ML problems
 - Node-level task: "How many units of Product 1 will User B purchase in Order 3?"



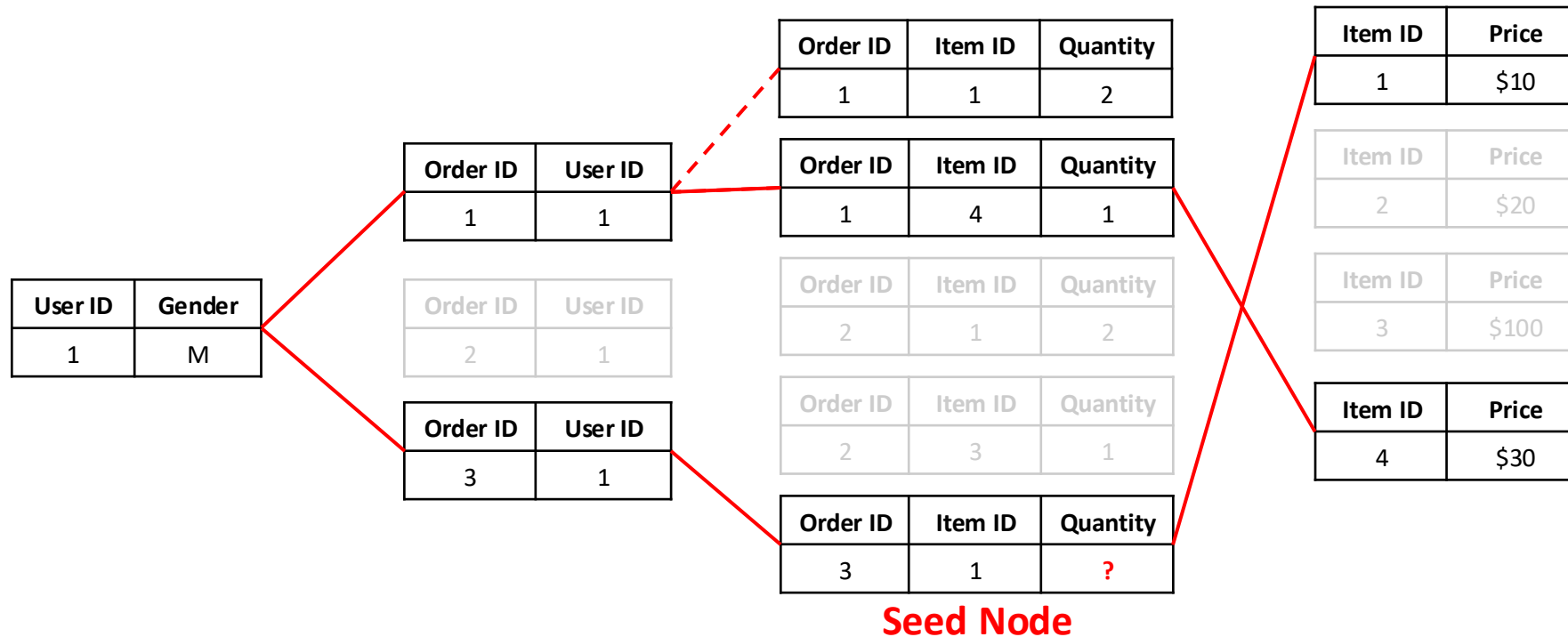
Graph Sampling (1)

- Centering on seed node, always include F->P keys



Graph Sampling (2)

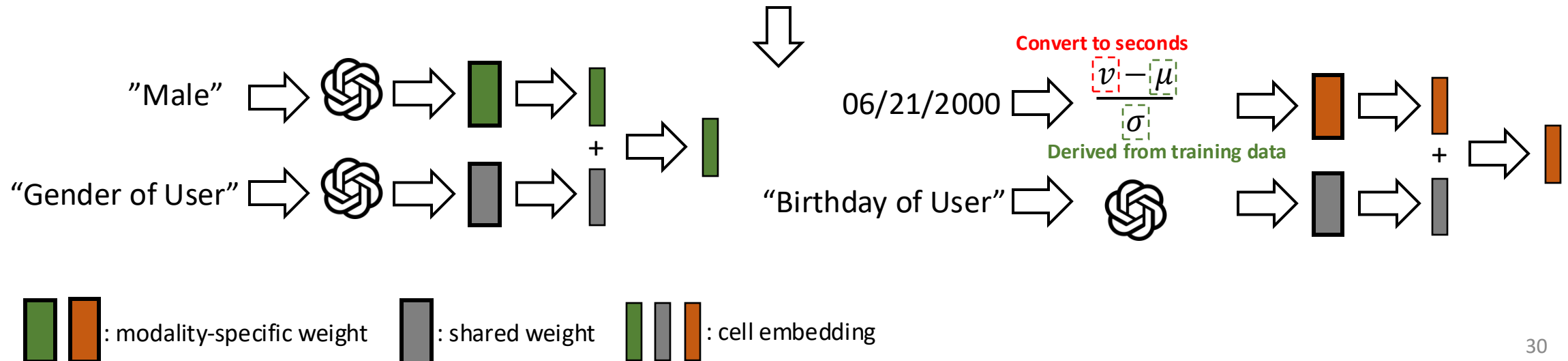
- Subsample child rows (P->F keys)
 - Take sampled size = 1 as an example



Input Representation (1)

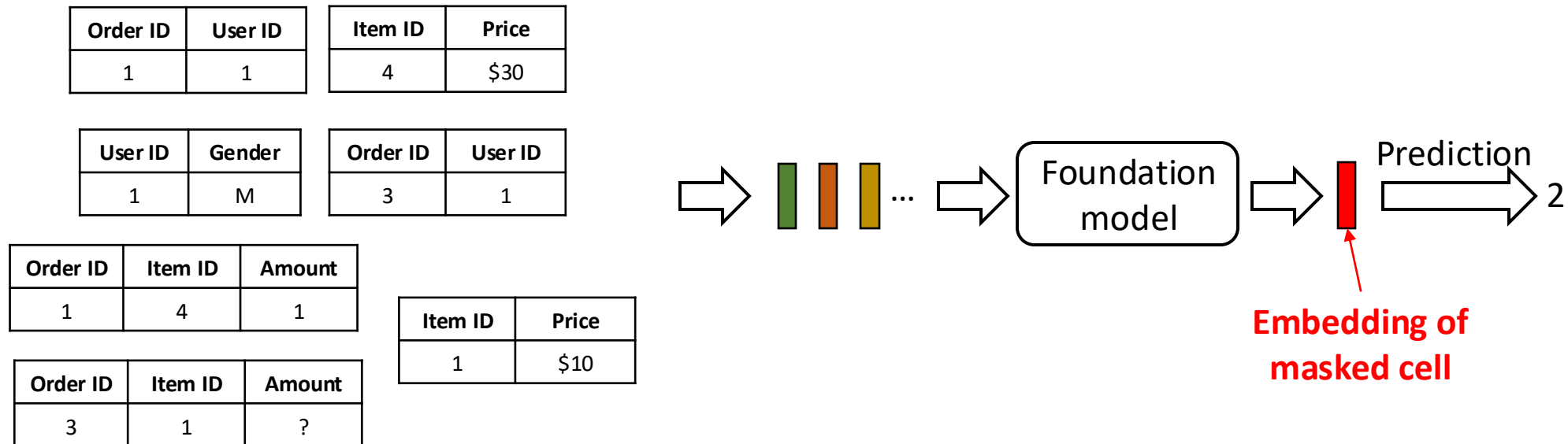
- Each cell in a row is represented as (value, column, table)
 - Value will be embedded with modality-specific encoder
 - Numerical value is first normalized based on calculated mean and std
 - Column and table information are injected via “<column_name> of <table_name>”

User ID	Gender	Birthday
1	Male	06/21/2000



Input Representation (2)

- All cell embeddings form an input sequences to foundation model
 - The token order is ignored



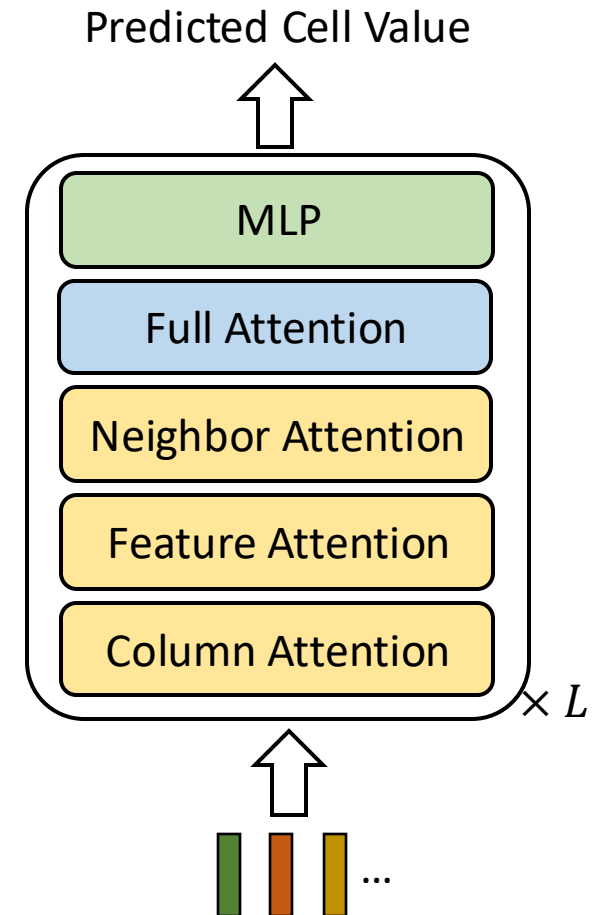
Architecture (1)

- The architecture includes four attention mechanisms
 - They all apply **Scaled Dot-Product Attention** backbone with different **Attention Resolution** (defined by specific masks)

$$\text{SoftMax} \left(\frac{\mathbf{QK}^T \odot \mathbf{M}}{\sqrt{d}} \right) \mathbf{V}$$

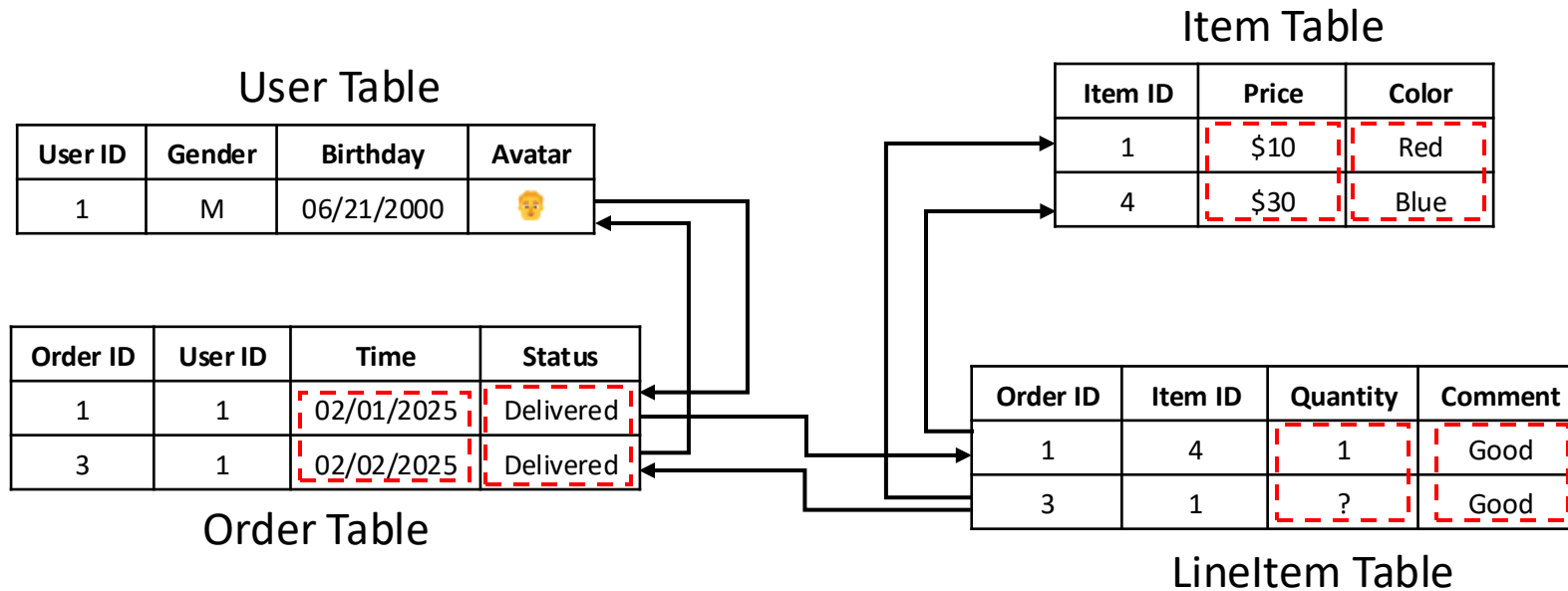
Attention Mask

- Column Attention: Cells attend to others in the same column
- Feature Attention: Cells attend to all others in the same row
- Neighbor Attention: Cells attend to their child rows (P->F keys)
- Full Attention: All tokens attend to each other



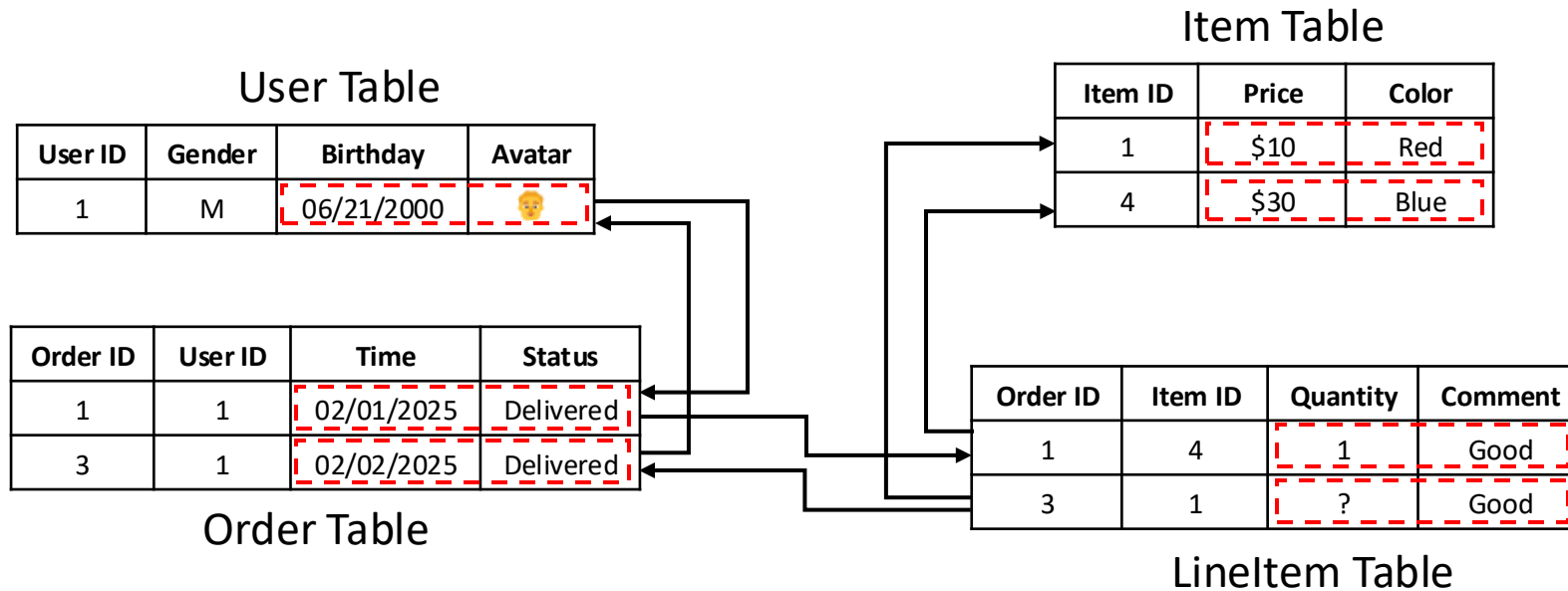
Architecture (2)

- Column Attention: Cells attend to others in the same column
- Cells within the same will perform attention to each other



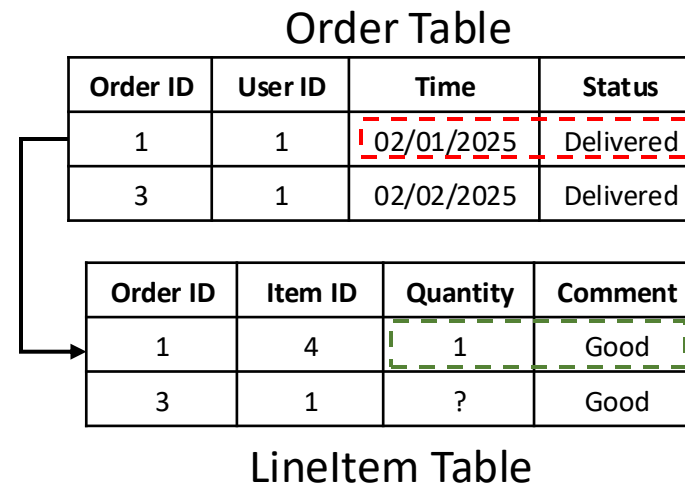
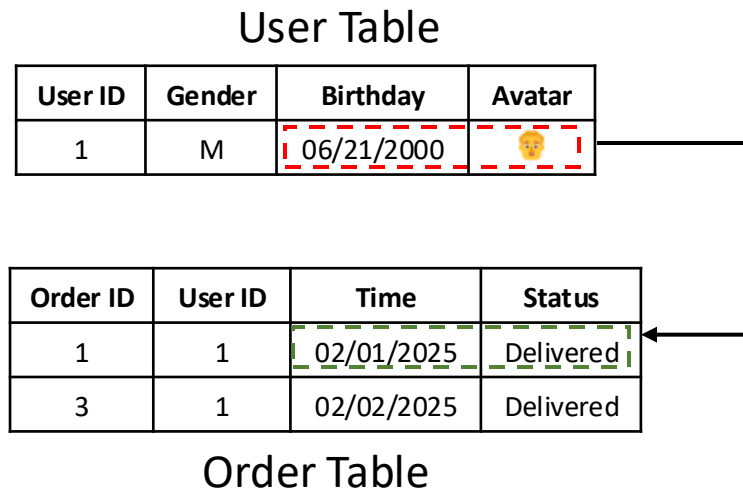
Architecture (3)

- Feature Attention: Cells attend to all others in the same row
- Cells within the same will perform attention to each other



Architecture (4)

- Neighbor Attention: Cells attend to their child rows (P->F)
- For the rows connecting by P->F, all cells within (query) will attend to the cells within (key)



Pre-training Pipeline: Masked Cell Prediction

- Workflow
 - Randomly sample a **seed cell** from the database and **mask** its original value
 - Sampling a **subgraph** centered around the seed node
 - Map raw cell values and metadata into a high-dimensional **embedding space**
 - **Flatten** the structural subgraph into a unified sequence of cell-level tokens
 - Optimize the model by reconstructing the **masked value**
- 7 Datasets are applied for pretraining
 - Covering E-commerce, Social, Sports, and Medical domains

Zero-Shot Transfer Results

- Tasks: 10 diverse binary classification challenges
 - The Metric (AUROC): Measures the model's ability to distinguish between classes (e.g., Churn vs. Stay). Higher is better.

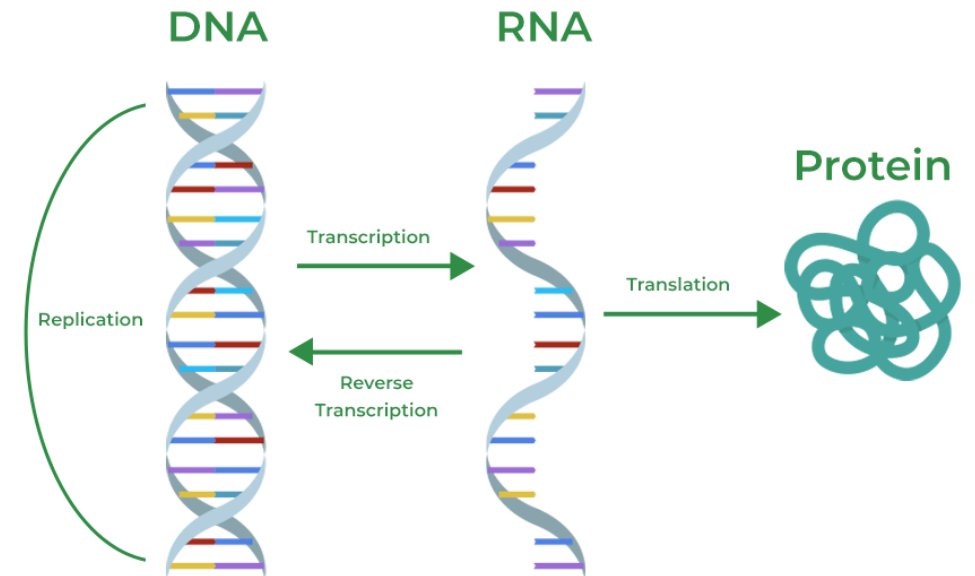
Target dataset $\in$ pretraining? $\rightarrow$		Maybe			w/o finetuning			w/ finetuning	
					No			Yes	
Dataset $\downarrow$	Task $\downarrow$	Gemma	Gemma	Gemma	Entity Mean	Griffin	RT (ours)	Griffin	RT (ours)
Parameter count $\rightarrow$		4B	12B	27B	0	22M	22M	22M	22M
rel-amazon	item-churn	62.1	55.0	42.1	73.0	69.0	70.2	71.9	74.3
rel-amazon	user-churn	58.1	54.7	50.5	64.4	62.3	63.9	64.1	65.2
rel-avito	user-clicks	54.5	59.5	59.8	44.7	45.9	59.5	45.9	60.8
rel-avito	user-visits	60.1	57.9	62.7	60.7	60.7	61.8	62.2	62.6
rel-fl	driver-dnf	56.2	54.6	75.8	75.4	57.7	82.0	57.7	82.0
rel-fl	driver-top3	84.6	90.5	91.4	85.0	82.5	89.1	81.8	89.3
rel-hm	user-churn	59.8	47.1	48.7	64.4	60.2	62.8	60.4	63.1
rel-stack	user-badge	79.1	79.8	80.0	66.2	73.5	80.0	82.3	83.6
rel-stack	user-engage	65.9	67.8	78.0	83.5	77.5	77.1	89.4	87.8
rel-trial	study-out	52.6	57.4	57.2	50.0	51.0	54.5	57.2	60.1
Mean AUROC $\rightarrow$		63.3	62.4	64.6	66.7	64.0	70.1	67.3	72.9

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Genomics (1)

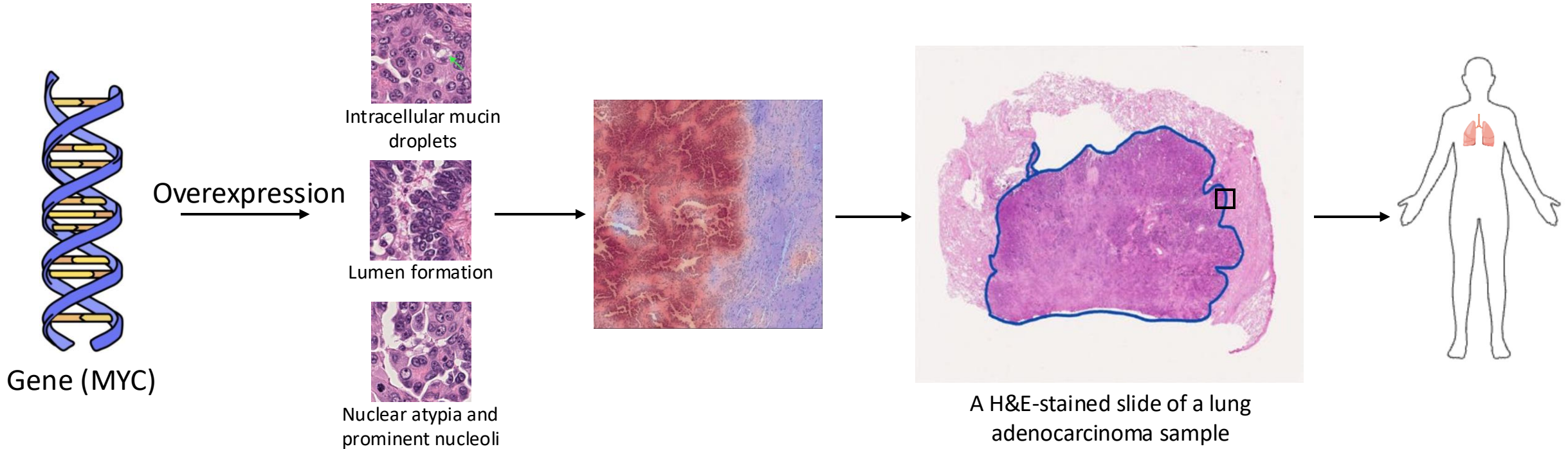
- The Central Dogma of Biology
 - DNA → RNA → Proteins
 - DNA contains genetic instructions
 - RNA carries these instructions to make proteins
 - Proteins perform cellular functions
- Gene Expression
 - Process of turning genetic information into functional products
 - Different cells express different genes



Blueprint -> Message -> Worker

Genomics (2)

- Gene Expression Level is a Critical Clinical Signal



Single-Cell RNA Sequencing (scRNA-seq)

- scRNAseq
 - Measures gene expression in individual cells
 - Reveals cellular diversity within tissues
 - Discovers new cell types
 - Tracks cell development and differentiation
- The Missing Piece
 - Cells are removed from their tissue
 - We lose information about where cells were located

	Cell1	Cell2	...	CellN
Gene1	3	2	.	13
Gene2	2	3	.	1
Gene3	1	14	.	18
...	.	.	.	.
...	.	.	.	.
...	.	.	.	.
GeneM	25	0	.	0

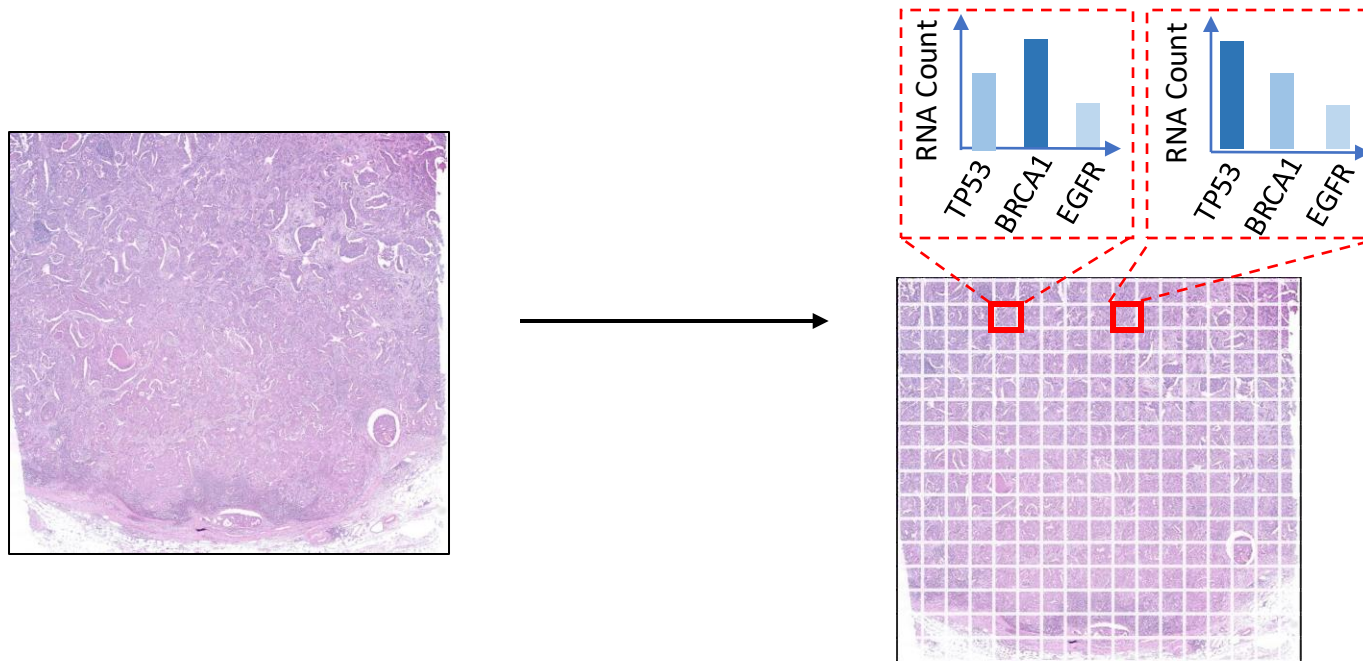
bulk: Average of all cells

scRNA-seq: Individual cells, no location

Spatial: Individual cells with location

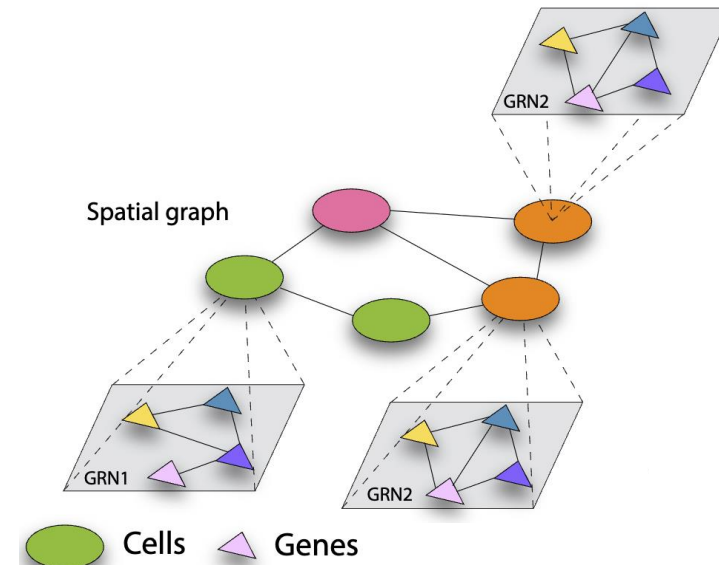
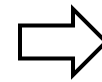
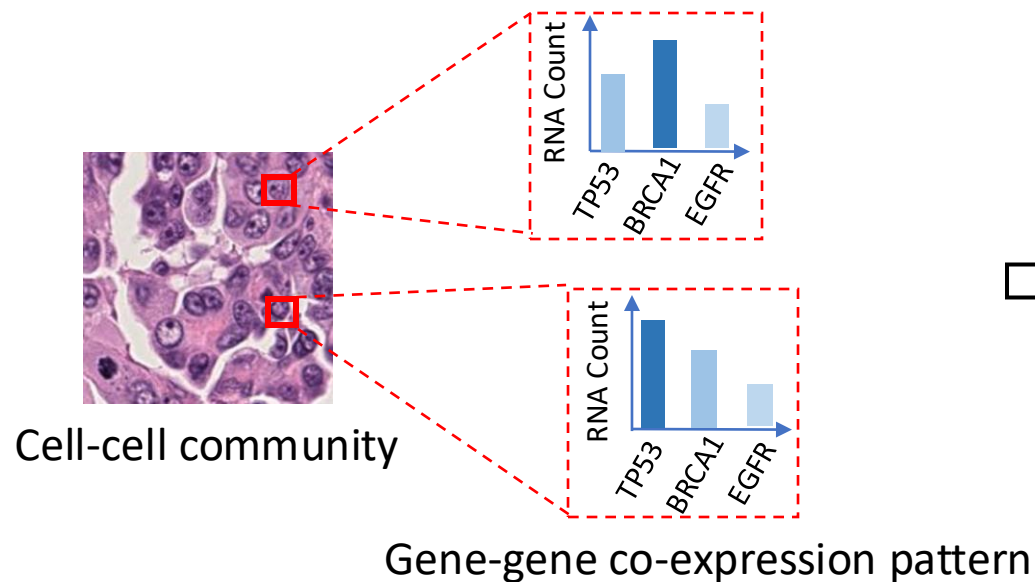
Spatial Transcriptomics: The Next Frontier

- What is Spatial Transcriptomics (ST)?
 - Measures gene expression while preserving spatial location
 - Like a molecular map of tissues
 - Shows both "what" (gene expression) and "where" (location)



Hierarchical Graph (1)

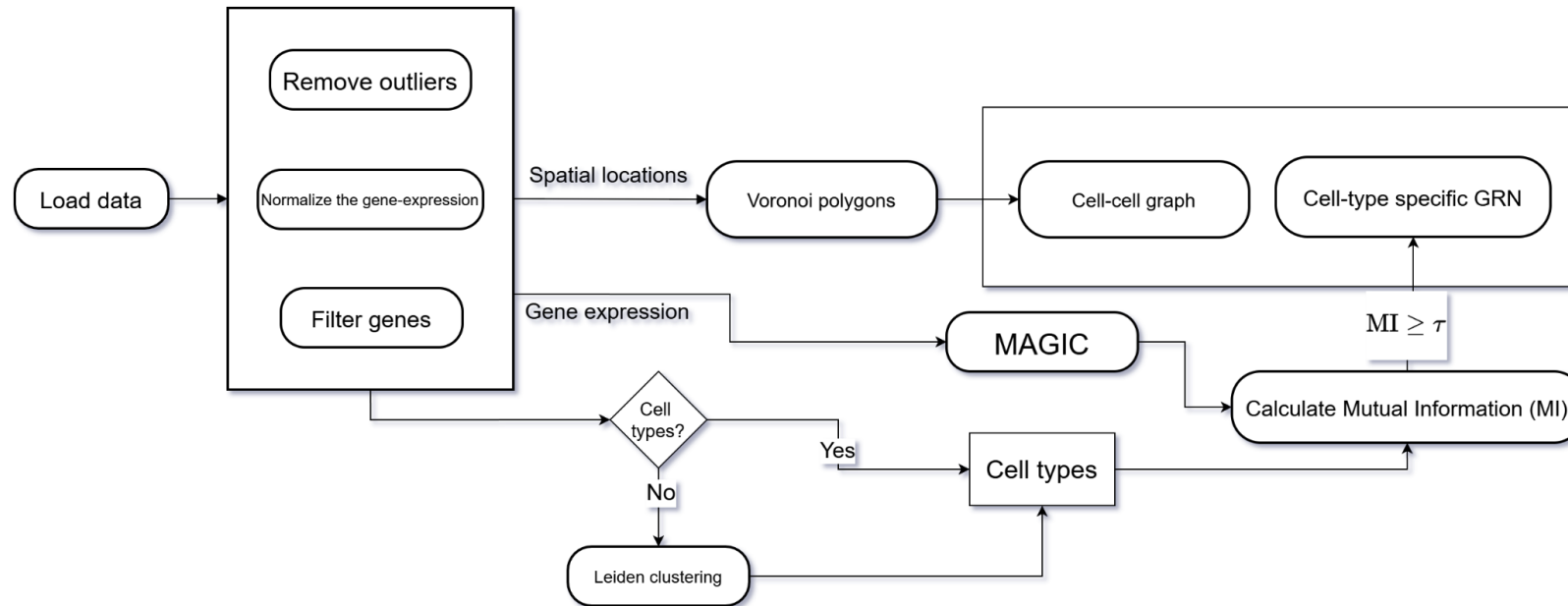
- Based on ST, we can have two levels of relational information
 - Cell-cell spatial interactions: neighboring cells influence each other's behavior
 - Gene-gene co-expressions: Groups of genes work together in coordinated programs to control cell function



Hierarchical Graph (2)

- Two-level hierarchy:
 - Cell level: Spatial graph capturing tissue organization
 - Gene level: Co-expression networks within each cell
 - Cross-level: Bidirectional message passing
- We convert each tissue into a set of graphs, consisting of
 - Spatial matrix $\mathbf{P} \in \mathbb{R}^{C \times 2}$, where each row gives 2D coordinates of a cell in the tissue
 - Gene expression matrix $\mathbf{X} \in \mathbb{R}^{C \times G}$, where each row contains expression values
 - Some meta data like gene identities, cell-types, etc.

Graph Construction



- Cell-Cell Graph: Voronoi adjacency from spatial coordinates
- Gene Co-expression Networks
 - Mutual information between gene pairs within each cell type
 - Edges for gene pairs with mutual information higher than a threshold

Architecture (1)

- Each layer has two main steps
 - Intra-level message passing and Cross-level message passing
- Repeat L times
 - $\mathbf{H}_c^{(l)} \in \mathbb{R}^{C \times d}$: representation for **cells**; $\mathbf{H}_g^{(l)} \in \mathbb{R}^{C \times G \times d}$: representation of **genes**

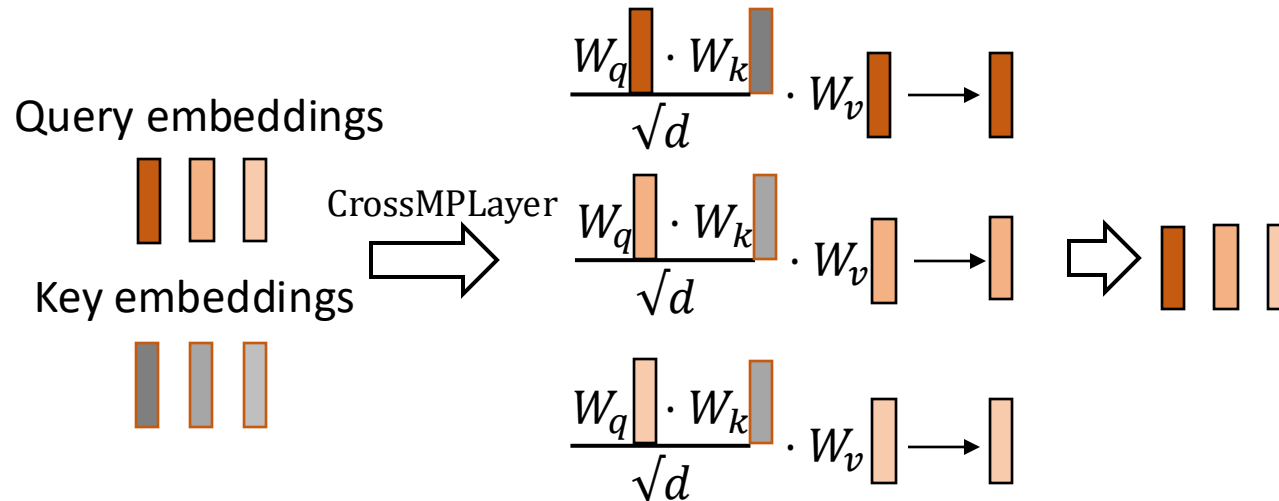
$$\mathbf{H}_c^{(l)}, \mathbf{H}_g^{(l)} = \text{Layer}(\mathbf{H}_c^{(l-1)}, \mathbf{H}_g^{(l-1)})$$

- Within layer, we first perform intra-level message passing
 - $\mathcal{E}$: edge set of **cell-cell** graph; $\mathcal{E}_k$: edge set of **gene-gene** graph at k -th **cell**

$$\begin{aligned}\hat{\mathbf{H}}_c^{(l)} &= \text{CellGraphTransformer}(\mathbf{H}_c^{(l-1)}, \mathcal{E}) \\ \hat{\mathbf{H}}_g^{(l)} &= \text{GeneGraphTransformer}(\mathbf{H}_g^{(l-1)}, \{\mathcal{E}_k\}_{k=0}^{|C|})\end{aligned}$$

Architecture (2)

- Then, we perform cross-level message passing (CrossMPLayer(query, key))



- Gene $\leftarrow$ Cell: Genes attend to their parent cell's spatial context
 - repeat means duplicating cell embedding to match all the corresponding gene embeddings

$$\mathbf{H}_g^{(l)} = \text{CrossMPLayer}(\hat{\mathbf{H}}_g^{(l)}, \text{repeat}(\hat{\mathbf{H}}_c^{(l)}))$$

- Cell $\leftarrow$ Gene: Cells aggregate their constituent gene representations
 - agg means aggregating all the gene embeddings within each cell to match cell embeddings

$$\mathbf{H}_c^{(l)} = \text{CrossMPLayer}(\hat{\mathbf{H}}_c^{(l)}, \text{agg}(\hat{\mathbf{H}}_g^{(l)}))$$

Objectives (1)

- Upon extracting cell and gene embeddings, the model is optimized via a multi-level contrastive learning framework
 - **Cell-Level Contrastive Loss**
 - **Positive Pairs:** Cells belonging to the *same* cell type.
 - **Negative Pairs:** Cells of *different* cell types.
 - **Gene-Level Contrastive Loss**
 - **Positive Pairs:** Genes co-expressed within the *same* cell.
 - **Negative Pairs:** Genes expressed across *different* cells.
 - **Cross-Modal (Cell-Gene) Alignment**
 - **Positive Pairs:** A cell and its *own* constituent gene embeddings.
 - **Negative Pairs:** A cell paired with gene embeddings from *other* cells.

Objectives (2)

- Alongside contrastive learning, HEIST employs a generative Masked Autoencoding (MAE) objective
 - Reconstruct the masked 1. the expression values of a subset of gene nodes, and 2. 2D coordinates of selected cells
- Combine contrastive and generative losses with a scalar
$$loss = \lambda L_{\text{contrastive}} + (1 - \lambda) L_{\text{generative}}$$

Experimental Setup

- Pre-training Data:
 - 22.3 million cells from 124 tissues across 15 organs
 - Technologies: MERFISH, Xenium (single-cell resolution)
 - Sources: 10x Genomics, Vizgen, Seattle Alzheimer's Brain Atlas
- Evaluation Tasks:
 - Cell clustering - Discover spatially-informed subpopulations
 - Cell type annotation - Classify cells into known types
 - Clinical outcome prediction - Tissue-level classification
 - Gene imputation - Recover missing expression values

Results

- Clinical Outcome Prediction
 - Predicting patient-level outcomes by aggregating the spatial and molecular embeddings of the entire tissue graph
 - Metric: AUROC
- Cell Type Annotation
 - Description: Classifying the specific identity or state of individual cells.
 - Metric: Accuracy

Clinical Outcome Prediction Results

Dataset Task	Skin Response	Placenta Condition	Colon Outcome	Colon Recurrence	Neck Outcome	Neck Recurrence
STAGATE	0.533	0.578	0.657	0.783	0.602	0.659
GraphST	0.644	0.659	0.828	0.645	0.582	0.683
scFoundation	0.500	0.601	0.713	0.787	0.678	0.689
CellPLM	0.580	0.682	0.744	0.801	0.681	0.667
scGPT-spatial	0.600	0.602	0.834	0.806	0.717	0.676
scGPT-spatial (aligned)	–	–	0.558	0.431	0.702	0.789
HEIST	0.866	0.769	0.861	0.887	0.835	0.929

Cell Type Annotation Results

Dataset	Lung MERFISH	Brain SEA-AD	Colon Charville	Neck UPMC	Neck DFCI
STAGATE	0.219	0.330	0.276	0.069	0.069
GraphST	0.408	0.230	0.368	0.062	0.058
scFoundation	0.150	0.250	0.322	0.022	0.041
CellPLM	0.504	0.670	0.476	0.056	0.057
scGPT-spatial	0.567	0.591	0.349	0.046	0.062
HEIST	0.513	0.995	0.534	0.283	0.112

Graph Foundation Models

- **Graph Foundation Models for Structure Encoding**
 - **WWW2026 - Towards A Universal Graph Structural Encoder**
- **Graph Foundation Models for Relational Database**
 - **ICLR2026 - Relational Transformer: Toward Zero-Shot Foundation Models for Relational Data**
- **Graph Foundation Models on Biology**
 - **ICLR2026-HEIST: A Graph Foundation Model for Spatial Transcriptomics and Proteomics Data**